

Group 5

Vision-Based Localization for Eye Snake robots

A Sim-to-Real Benchmark and Adaptation Study

CIS II Project Checkpoint Presentation by Chuntung Zhuang
Mentored by Prof. Iulian Iordachita, Adnan Munawar, Mojtaba Esfandiari



Presentation Outline

01 Project Summary

Problem motivation and approach overview



02 Technical Progress

Data pipeline, benchmark results, and key findings



03 Management

Timeline, challenges encountered, and next steps



Problem & Motivation

The Challenge

- Micro-continuum surgical robots enable dexterous manipulation in confined spaces like the eye
- **Tendon hysteresis and compliance** make encoder-based localization unreliable
- Sub-mm precision required — errors cause tissue damage in retinal surgery
- **Labeled real data is extremely scarce** and expensive to collect

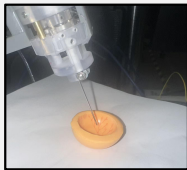
Our Approach

- Vision-based localization — bypass hysteresis modeling using camera feedback
- AMBF simulation toolkit for realistic and scalable synthetic data with domain randomization

Goal: Achieve sub-millimeter localization

Step1: Data Collection

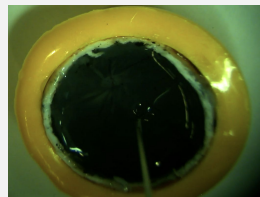
Real



a) Micro-Continuum Robot



b) Data Collection Setup

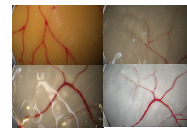


c) Real and Phantom Porcine Eye

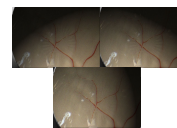
Sim



a) Simulation Toolkit Model

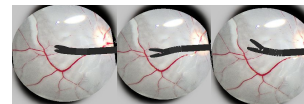


b) Diverse Retinal Background

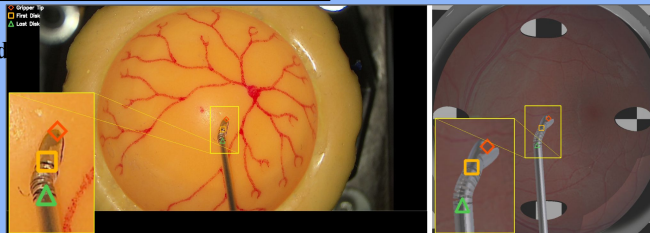


c) Configurable Illumination Conditions

d) Controlling the Robot to bend in increasing angle



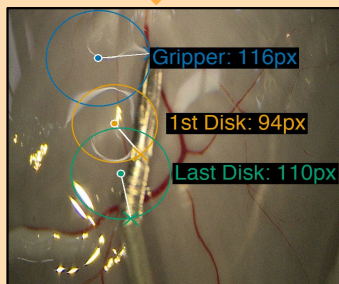
d) RosBag Video Extraction and Manual Keypoint Annotations



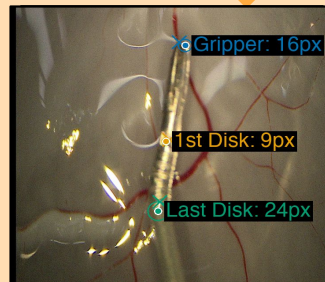
e) Projecting 3D keypoint positions onto UV coordinate for keypoint annotations

Step2: Training

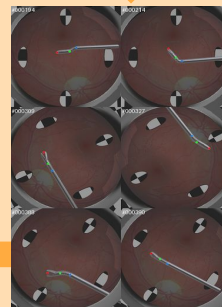
e) Real-only training from scratch



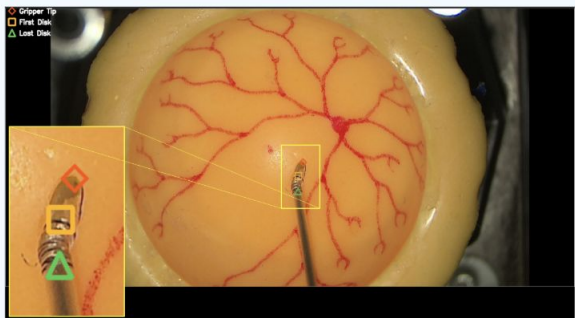
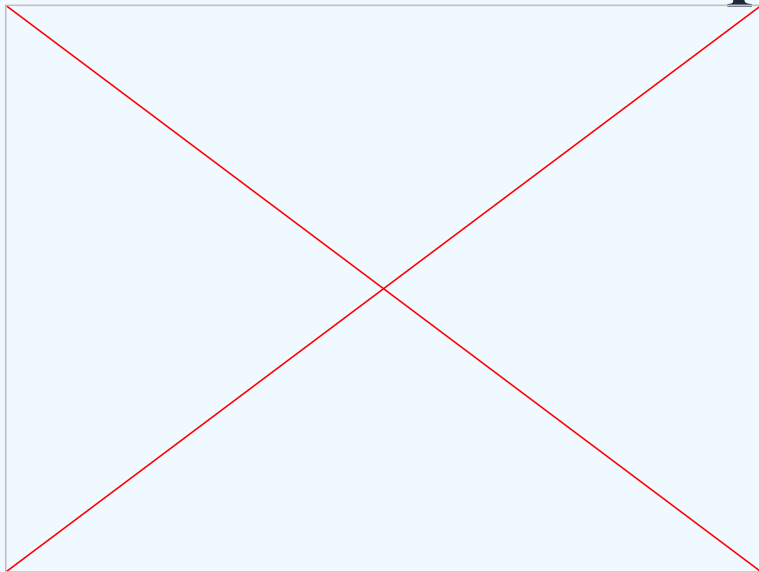
VS.



f) Curate Full simulation Dataset
g) Supervised pretraining on simulation data
h) Fine-tuning on real data



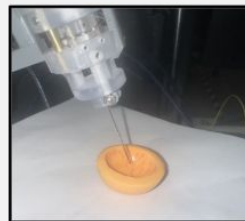
Dual-Domain Data Pipeline



Real Data

- Real and Phantom Pig eyes under Leica microscope
- 6 video sequences from porcine eyes
- 400 frames extracted, 217 fully annotated
- 3 keypoints: gripper, 1st disk, last disk
- Manual annotation + consistency review

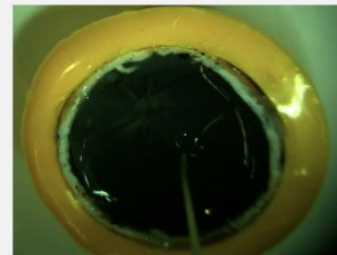
Real



a) Micro-Continuum Robot



b) Data Collection Setup



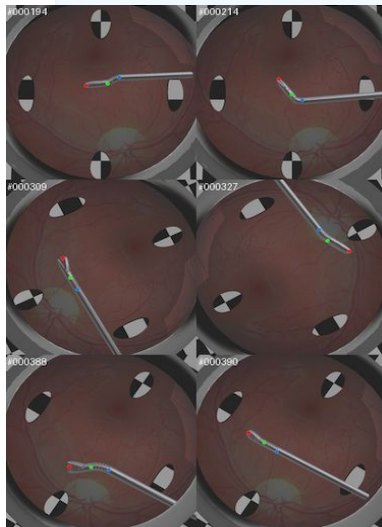
c) Real and Phantom Porcine Eye

Dual-Domain Data Pipeline



Simulation Data

- AMBF-based simulation toolkit
- 300 robot configurations sampled
- 4 retinal textures $\times$ 3 lighting = 12 visual conditions
- Auto 2D keypoint labels via 3D $\rightarrow$ 2D projection



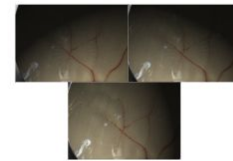
Sim



a) Simulation Toolkit Model

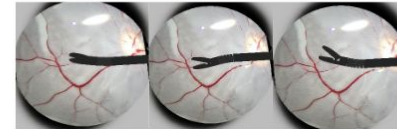


b) Diverse Retinal Background

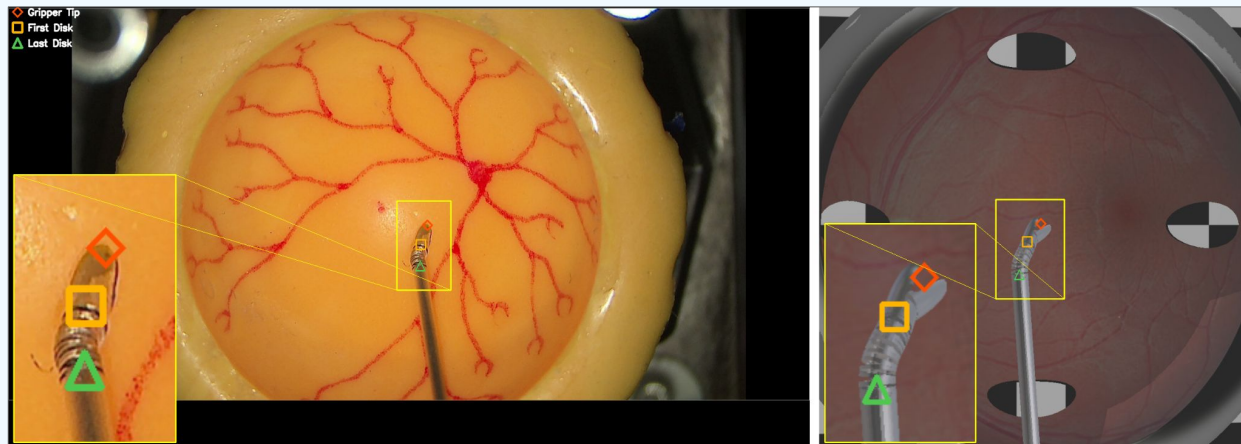
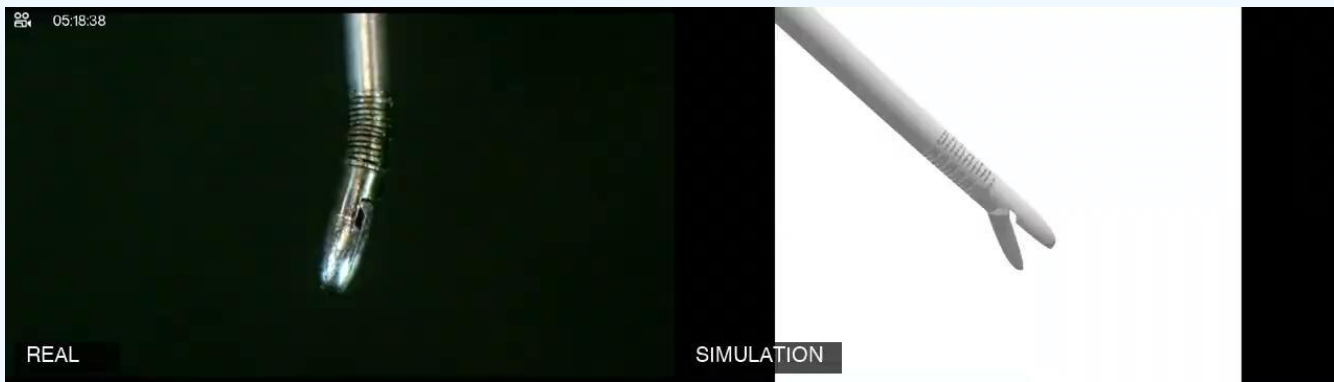


c) Configurable Illumination Conditions

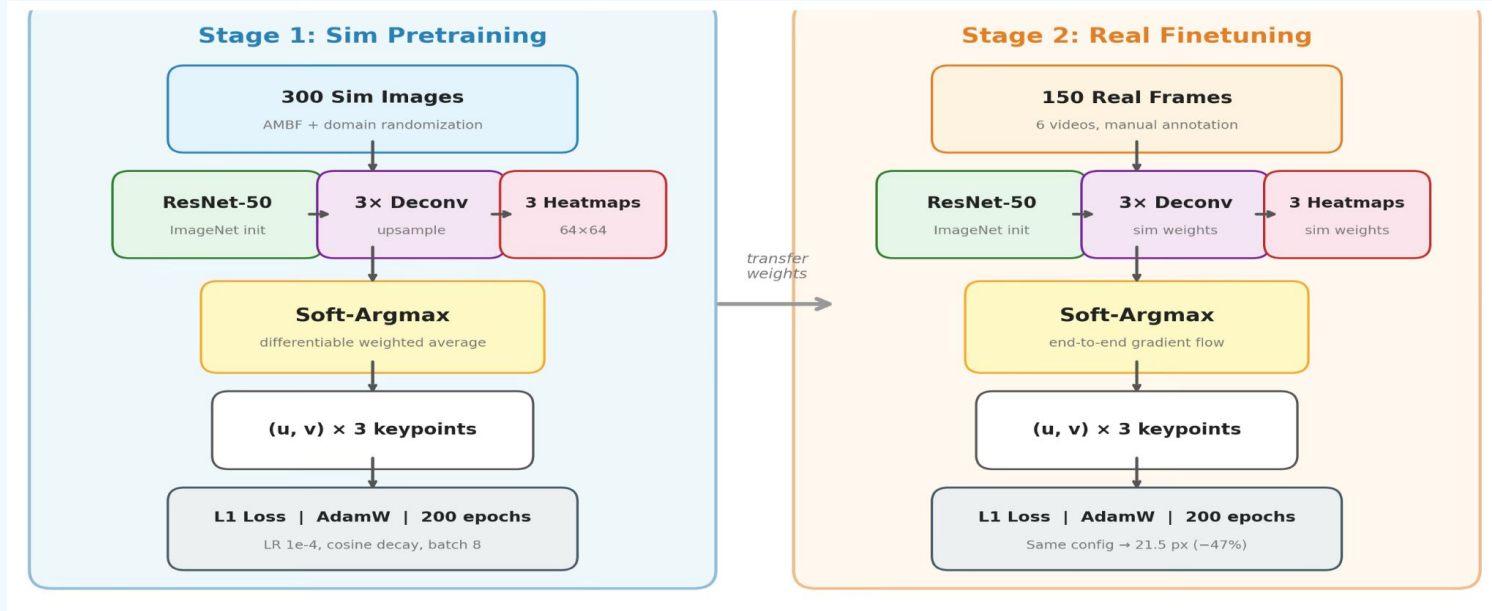
d) Controlling the Robot to bend in increasing angle



Dual-Domain Data Pipeline



Training: Integral Regression



Compute & Dependencies

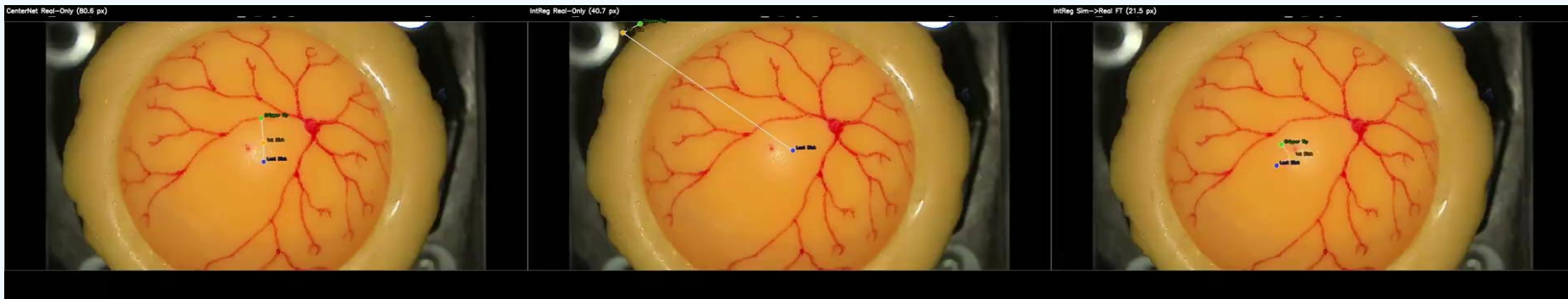
Input resolution: 512×512 (squashed from 1920×1080)

Batch size: 8

GPU: NVIDIA RTX 5090 (rented on Vast.ai, self-funded)

Framework: PyTorch + torchvision

Demonstrations



We compare three models on the same real surgery video (Video 3) running at 30 fps.

Left: CenterNet with real-only training (80.6 px mean error).

Center: Integral Regression with real-only training (40.7 px mean error).

Right: Integral Regression with sim-to-real finetuning (21.5 px mean error)

Timeline & Plan Updates

Completed

Deliverable

Details

- | | |
|--|---|
| ✓ Simulation data pipeline | AMBF + domain randomization, 300 labeled sim images |
| ✓ Real data collection & annotation | 6 surgery videos, 200 frames, 3 keypoints each |
| ✓ Camera intrinsics calibration | Calibrated the cameras to collected videos |
| ✓ ROS capture | Deployed on LCSR Desktop with ROS2BAG |
| ✓ Working Sim-to-real finetuned model | Integral Regression, 21.5 px mean error |
| ✓ Real-time inference demo | 30 fps labeled video, 3-model comparison demo |
| ✓ Sim to Real Gap Analysis | compare the performance between sim to real and real only performance |

Timeline & Plan Updates

Expected and Maximum deliverable

● improve to > 20px of error (before April 20th)

- applying ML techniques like multi-model resolution ensembling
- Ablation comparing ensemble vs. single-best-resolution baseline

■ Stereo video collection (before April 25th)

- Collect synchronized stereo sequences via dual Zeiss microscope cameras
- Prerequisite: ROS 2 bag recording pipeline operational
- Risk: primary camera has slight defocus; backup camera confirmed available by mentor
- Validation: epipolar error below threshold (currently being reduced via calibration)

my mentor confirmed in this Monday's meeting that there might be another camera

■ Dense Stereo depth map generated (before April 30th)

- Foundation Stereo or COLMAP for dense depth maps
- Deliverable: per-frame depth maps with qualitative & quantitative evaluation

■ 3D localization on stereo data — target: early May (begins once de

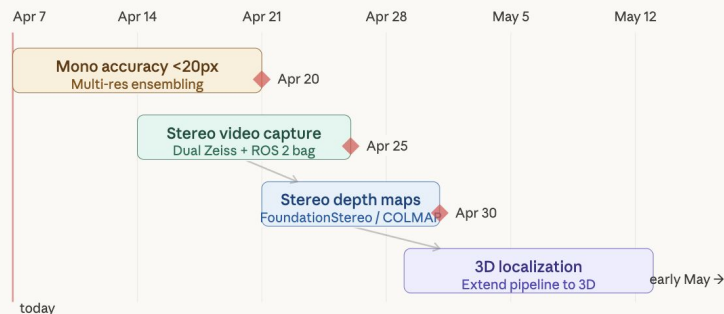
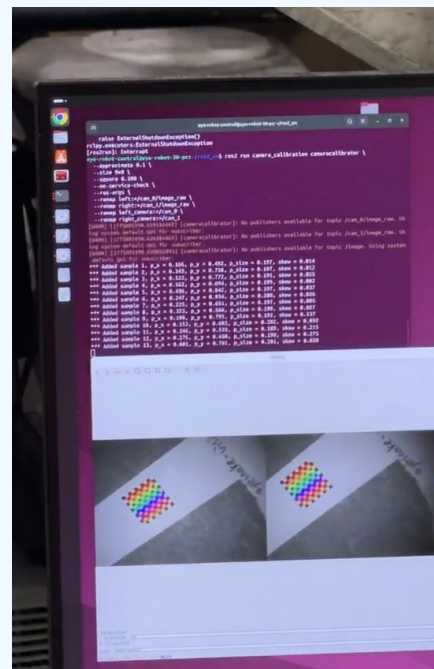
- Extend localization pipeline from 2D to 3D using estimated depth
- Compare 3D error metrics against mono baseline

Time Arrangement:

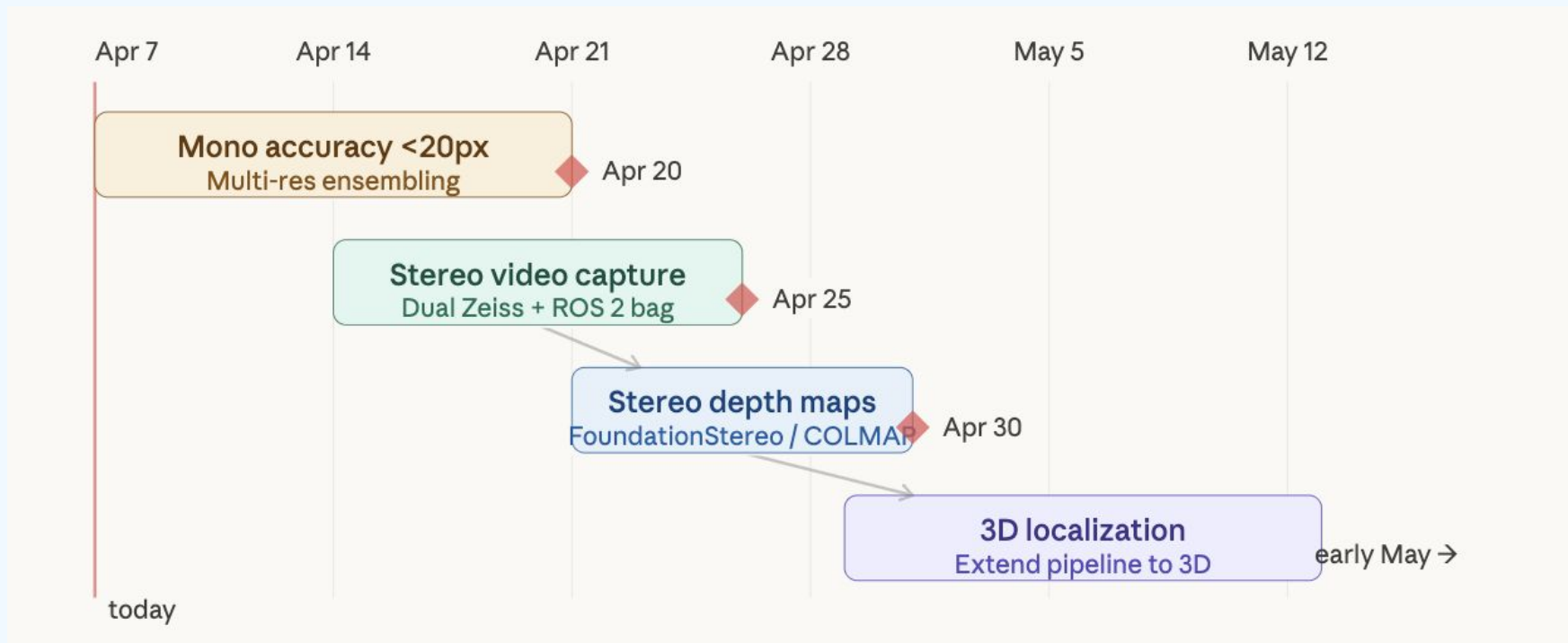
Mentor meetings: Mon & Wed

Lab experiments (LCSR): $\geq 2 \times$ /week

Documentation: biweekly PowerPoint update → wiki, presented each m



Timeline & Plan Updates



Thank You

Questions?